5 **(a)** When the resistance of the variable resistor is increased, it results in a decrease in current in the large loop. This results in a decrease in the magnetic field in the region enclosed by it. 1

As such there is a decrease in magnetic flux linking through the small loop. By Faraday's law, this result in an induced e.m.f., which causes a current to flow as the loop is a complete circuit. 1

(b) Current in outer loop is anti-clockwise and the magnetic field it encloses is out of the page. 1

Since this magnetic field is decreasing, by Lenz's law, the induced current will flow anticlockwise too, so as to produce its own field into the page to oppose the decrease in flux. 1

(c) $B = \frac{\mu_o I}{2r}$ (from formula list)

$= \frac{\mu_o (2.0)}{2\left(\frac{0.200}{2}\right)}$

$= 1.2566 \times 10^{-5} = 1.26 \times 10^{-5}$ T 1

(d) **(i)** Average induced e.m.f., $\langle\varepsilon\rangle = \Delta\phi / \Delta t$

$= (\Delta B)A / \Delta t$

$= \frac{\left[\frac{\mu_o (2.0 - 1.0)}{2(0.10)} \times \pi (0.015)^2\right]}{0.25}$

$= 1.7765 \times 10^{-8}$ V 2

Average induced current, $\langle I\rangle = \langle\varepsilon\rangle / R$

$= 1.7765 \times 10^{-8} / 1.5$ 1

$= 1.18 \times 10^{-8}$ A

(ii) The magnetic flux density calculated in **(c)** is actually the flux density at the centre of the large coil. In the calculation for **(d)(i)**, we assumed that the magnetic field enclosed by the inner coil is at this value throughout. 1

(e) Even though there is still an induced emf (by Faraday's law), 1
no current flows because the resistance of an insulator is very high (infinite). 1

[Total: 11]

6 (a) (i) A packet (or quantum) of electromagnetic radiation energy; 1